

SKILLS

- **Code** – Rust | Python | JavaScript | Typescript | React | C | Pytorch | C# | C++ | Java | bash | Git | CDK | SQL | ARM & Cross-tooling. | Linux kernel & iOS.
- **Tools** – DynamoDB, S3, APIGateway, CloudWatch, CDK, CloudFormation, Lambda, IAM, etc | TensorFlow & Keras | Kubernetes | System Performance.
- **Logic** – REST, Serverless & Network | Async, load-balancing & low-latency | Full-stack | Advanced Math & Statistics | ML, Deep-Learning, NLP & GenAI.
- **Security** – CVE-lifecycle- CVSS, NIST, CISA, Mitre, Debian, Redhat, FIPS | Threat Modeling & Remediation | Adversarial data & Mitigation | CISSP, CBK.

EXPERIENCE

A) Amazon Web Services (AWS) – Software Developer | EC2, Amazon Linux Threat Mgmt. | Seattle, WA. **AUG 22'–Now (3 yrs., 6 mos.)**

1. **Security & Operations** : distros - ALI2, AL1, AL2, AL2023 - (incl. AL2023-desktop & all EC-2 system-packages).
 - ① **Linux CVE Management & Response** – Managed Linux & kernel CVE detection with a strict 2-day SLA, handling 1,475+ CVEs, including Sev-2 incidents and remediation planning.
 - ② **Vulnerability Patching** – Fixed 15+ security vulnerabilities in Amazon Linux packages, some requiring FIPS compliance.
 - ③ **Knowledge Sharing & Documentation** – Maintained critical runbooks and mentored new developers.
2. **Design & Development** : AWS - Linux docx. [docs.aws.amazon.com/linux ]
 - ① **Full-stack & Operational features** – On top of triaging, mitigating, resolving complex bugs & bottlenecks across 2 legacy services; I pushed 12+ React features for a 3rd, new edge-compute APIGW service with role-specific access, to collaboratively deliver a high-frequency, data-intensive UI to provide real-time CVE status + related data; all including unit, integration & regression test coverage;
 - ② **Cloud Development & Infrastructure Migration** – Migrated Rust-based client infra coupled with Py Lambdas, across services & engineered nested CloudWatch alarms for edge-compute based rate-limiting, request throttling, etc with robust, integrated DevOps.
 - ③ **Networking & Concurrency Architecture** – Built privacy-enhanced RESTful features & secure client-side behaviors across distributed AWS infrastructure across services (DynamoDB, IAM, S3, APIGW, CloudFormation, etc). This includes several dense multithreaded + async workflows (Rust, Py, Js), each optimized for resilience against all classes of network & edge-case failures.
3. **Analytics & Automation** : Amazon Linux Security Centre CVE advisories (ALCE) [alas.aws.amazon.com/faq.html ]
 - ① **Automated CVE Evaluations** – Designed AI-driven CVE scoring and recommender systems, integrating AmazonQ-based chatbots to reduce zero-day SLA targets from two days to near real-time.
 - ② **Security focused NLP & Statistics** – Built an automated CVE impact analysis algorithm, applying NLP & time-series analytics to enhance security risk assessment and reduce manual workloads for customers.
 - ③ **Scripting** – Automated repetitive steps via custom scripts, saving approximately one week per month in manual overhead.

B) BP Logix [bplogix.com] – Software Development Intern | San Diego, CA. **JUN 21'–SEP 21' (4 mos.)**

1. **Version Control Migration**: Led the transition from TFVC to Git, creating a comprehensive run-book for seamless replication.
2. **DevOps & Agile Exposure**: Delivered two bug fixes in C# using VScode, engaged in Scrum processes, enhancing understanding of Agile development, KPIs & release goals.

C) Cleo Communications [cleo.com] – Software Developer | Bengaluru, IN. **SUMMER 19' & FEB 20'–DEC 20' (3 + 11 mos.)**

1. **ML-Powered Data Transformation**: Built a pipeline to map features across disparate datasets, evaluating RNNs, Logistic Regression, and Fuzzy Logic (which performed best).
2. **Product Testing & Debugging**: Conducted cross-service unit testing on CIC Clarify (Node & Cluster); documented bugs via Postman.

D) Samsung R&D India – Research Intern | Bengaluru, IN. **MAR 19'–NOV 19' (8 mos.)**

1. **Web Crawling & Data Extraction**: Developed subject-specific crawlers to extract social media data for research and analysis.

PUBLICATIONS & RESEARCH PROJECTS — [approx. 50 Cites from 2021 to 2025]

[Google Scholar](https://scholar.google.com/citations?user=57221594595) | [Scopus Author ID: 57221594595](https://orcid.org/0009-0000-8197-704) | [ORCID record: 0009-0000-8197-704](https://orcid.org/0009-0000-8197-704) | [IEEE Author ID: 37088644371](https://orcid.org/37088644371) | [ResearchGate](https://researchgate.net/profile/Keerthana-Purushotham)

→ Reviewed 8+ IEEE papers in 2025. | → Member of [CVE.org Working Groups](https://cve.org/WorkingGroups).

1. **Text To Video App** – Async API with the Genmo Mochi-1 model hosted on 8xH100 GPU Kubernetes work nodes. | [LinkedIn](#) | [GitHub](#) – WIP.
2. **Accuracy Is Not Enough** – Confusion Matrix Metrics That Actually Work in CVE Impact Prediction. | [LinkedIn](#) | [Substack](#) | [Medium](#)
3. Cloud Storage Security Risks, Practices & Measures: A review. | 40 cites | [IEEE Xplore](#) – Jan 1, 2020.
4. Image de-noising using Auto-encoders & Spatial Filters for Gaussian Noise. | 7 cites | [IEEE Xplore](#) – Mar 15, 2021 | [GitHub](#).
5. Context-Based Filtering of Conversational Data. | 2 cites | [ACL](#) – May 22, 2021 | [GitHub](#).

EDUCATION

A) University of California, San Diego (UCSD) – M.S. Computer Science | GPA 3.83/4.00 | San Diego, CA. **DEC 20'–JUN 22' (2 yrs.)**

[AI-&-Systems] – Courses : Algorithm Design | adv. Data Structures | Operating Systems | Computer Organization & System Programming (Arm) | OS Security | ML-algorithms | Probabilistic reasoning | Recommendation Systems & Web mining | adv. Statistical NLP | Structured Prediction in NLP | Ethics, society & Data Science.

1. Graduate Teaching Assistant : Mar 21 – Jun 22', 16 mos.

- ① [Spring 22'] **Advanced Data Structures** | CSE 100 – class of 300+ students with Paul Cao & Dr. Debashis Sahoo.
- ② [Winter 22'] **Graduate Algorithm Design & Analysis** | CSE 202 – class of 500+ students with Dr. Russel Impagliazzo.
- ③ [Fall 21'] **Graduate Algorithm Design & Analysis** | CSE 202 – class of 500+ students with Dr. Ramamohan Paturi.
- ④ [Spring 21'] **Computer Organization & Systems Programming** | CSE 30 – class of 500+ students with Bryan Chin.

2. Grace Hopper Scholar : 2021–2022, 1 year

3. Projects & Graduate Course Work : Jan 21'–Jun 21', 6 mos.

- ① NLP | **Neural CRF for Constituency Parsing** – [github](#)  – CKY-based constituency parser on PTB data, to optimize the partition fn on GPUs.
- ② NLP | **Neural CRF NER Tagger** – [github](#)  – Bi-LSTM-CRF models for Named Entity Recognition (NER), beating baseline in sequence labeling.
- ③ OS | **Ubuntu System Performance** – Analyzed CPU, memory, scheduling, networking & FS for metrics like efficiency, latency etc.

4. Graduate Research Apprentice : Jan 22'–Jun 22', 6 mos. | NLP & Recommender Systems Lab advised by Dr. Julian McAuley

- ① **NLP in Finance & Stats** – Researched stock prediction via text inference & trend analysis; specialized in equities, options & futures.

B) Ramaiah Institute of Technology (MSRIT) – B.E. Computer Science | GPA 9/10 | Bengaluru, IN. **AUG 16'–AUG 20' (4 yrs.)**

1. **Technical Awards** : ① Best Final Year Project 2020 MSRIT | ② Samsung PRISM Research Program Finalist 2020.

2. **Certifications** : ① Docker: Microfocus | ② Deep Learning for CV: NVIDIA | ③ Data Sc. for Engineers: NPTEL | ④ IoT - Raspberry Pi: Vocational Course

C) Other – Schooling | Bengaluru, IN. **JUN 01'–APR 16' (15 yrs.)**

1. **Grades** : ① [Sophia High School] 10th ICSE Boards: 92% | ② [Deeksha Integrated IIT Coaching] 12th Boards: 90%
2. **Awards & Activities** : MUN, debate, theater, school 1st rank & state top 50 several times between 2010 & 2015 across the annually-conducted, international Olympiads organized by the Science Olympiad Foundation (SOF), athlete in field / track events. Received the following medals -
 - ① IMO (Math): 3 Gold, 2 Silver | ② ISO (Science): 4 Gold, 1 Silver | ③ IEO (English): 1 Silver, 1 Bronze.